The Reciprocal Resonance–Fredholm Dictionary

Reconnecting Rank-Growing Clouds to the Fixed-Noise Determinant

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Abstract

The rank-growing atlas records operator resonances λ, while a Fredholm determinant in spectral variable z vanishes at $z = 1/\lambda$. This paper makes that change of variable explicit and prevents the direct resonance divisor rejected in RH-225 from being confused with the correct determinant divisor.

For a finite resonance multiset Λ of size n and characteristic polynomial $p_\Lambda(w) = \prod_{\lambda\in\Lambda}(w-\lambda)$,

$$\prod_{\lambda\in\Lambda}(1 - z\lambda) = z^n p_\Lambda(1/z).$$

Second regularization multiplies each factor by $e^{z\lambda}$ and therefore does not change the zeros. Conjugate resonance clouds produce conjugate reciprocal-zero clouds and real entire products.

Across the 32 physical endpoints, the identity is evaluated on 3,072 complex grid points with maximum error 9.16×10^{-15}. Reciprocal moduli range from 1.15129 to 13.54281, and the maximum scale-free zero-factor residual is 2.29×10^{-16}. Direct evaluation of the full regularized product at the outer zeros is numerically unstable, reaching a raw residual of 6.51×10^4; this illustrates why factorwise or logarithmic diagnostics are required.

The infinite fixed-positive-noise Hilbert–Schmidt determinant is already a rigorous result of RH-7. The contribution here is the exact dictionary from the new finite clouds to that determinant. No locally uniform small-noise limit or self-adjoint spectral interpretation is asserted.

1 Correcting the spectral variable

RH-225 proves that a tight rank-growing cloud cannot itself be a locally finite zero divisor [5]. That result concerns the points λ_j in the operator spectrum. A Fredholm determinant is instead a function of a spectral parameter z:

$$D_A(z) = \det(I - zA). \tag{1}$$

If A has nonzero eigenvalues λ_j, the zeros of (1) are their reciprocals. This distinction is standard but decisive for the current route.

2 Finite polynomial dictionary

Let

$$p_\Lambda(w) = \prod_{j=1}^{n}(w - \lambda_j). \tag{2}$$

Proposition 2.1 (Reciprocal polynomial identity). *For every $z \in \mathbb{C}$,*

$$F_\Lambda(z) := \prod_{j=1}^{n}(1 - z\lambda_j) = z^n p_\Lambda(1/z), \tag{3}$$

where the expression at $z = 0$ is interpreted by continuity. The finite zeros are exactly λ_j^{-1} for nonzero λ_j, with the same algebraic multiplicities.

Proof. For $z \neq 0$,

$$z^n p_\Lambda(1/z) = z^n \prod_j \frac{1 - z\lambda_j}{z} = \prod_j (1 - z\lambda_j).$$

Both sides equal one at $z = 0$. The factorization gives the zero statement. □

If Λ is the complete spectrum of a finite matrix A, then $F_\Lambda(z) = \det(I - zA)$. If it is a selected spectral cloud, F_Λ is the selected finite factor, not the determinant of the unresolved complement.

3 Second regularization

For a finite multiset define

$$F_{\Lambda,2}(z) = \prod_{j=1}^{n}(1 - z\lambda_j)e^{z\lambda_j}. \tag{4}$$

Corollary 3.1 (Regularization preserves the divisor). *The zeros of $F_{\Lambda,2}$ are exactly $\{\lambda_j^{-1}\}$ with algebraic multiplicity.*

Proof. The exponential factors are entire and nowhere zero. □

The logarithm normalized by $\log F_{\Lambda,2}(0) = 0$ is

$$\log F_{\Lambda,2}(z) = \sum_j \left[\log(1 - z\lambda_j) + z\lambda_j\right] = -\sum_{m\geq 2} \frac{z^m}{m} \sum_j \lambda_j^m \tag{5}$$

whenever $|z| \max_j |\lambda_j| < 1$. The missing linear term is the purpose of second regularization.

4 Conjugation symmetry

Proposition 4.1 (Real regularized factor). *If Λ is conjugate closed, then*

$$\overline{F_{\Lambda,2}(\overline{z})} = F_{\Lambda,2}(z).$$

In particular $F_{\Lambda,2}$ is real on the real axis and its reciprocal zero divisor is conjugate closed.

Proof. Conjugation permutes the factors in (4). □

Thus the shell-completion work of RH-222–RH-223 passes exactly to the Fredholm variable; no new pair matching is required [3, 4].

5 Fixed-noise infinite completion is inherited

For every fixed $\sigma > 0$, RH-7 proves that the centered folded Gaussian operator $\mathcal{N}_\sigma$ is Hilbert–Schmidt and constructs

$$\mathcal{D}_\sigma(z) = \det{}_2(I - z\mathcal{N}_\sigma) = \prod_j (1 - z\lambda_{\sigma,j})e^{z\lambda_{\sigma,j}}. \tag{6}$$

Its zeros are the reciprocal non-Perron resonances [2, 6]. Removing the negative parity mode as an additional finite factor gives the bulk determinant used by the later small-noise route.

Equation (6) is not reproved or claimed as new here. The rank-growing atlas samples its bulk resonance factors after the inherited Hardy scaling and Haar channel construction. What remains open is uniform control as $\sigma \downarrow 0$, not existence at fixed positive noise.

6 Numerical audit

For each endpoint, Proposition 2.1 is evaluated at 96 points: 24 equally spaced angles at radii $0.25, 0.5, 0.75, 1$. The archive also compares newly computed reciprocals with those stored by RH-222.

diagnostic	result
endpoint count	32
polynomial identity evaluations	3,072
maximum identity error	9.16×10^{-15}
maximum archived reciprocal error	floating zero
maximum reciprocal conjugacy error	binary64 scale
minimum reciprocal modulus	1.15129
maximum reciprocal modulus	13.54281
maximum zero-factor residual	2.29×10^{-16}

All reciprocal zeros lie outside the unit disk because all selected Hardy-scaled resonances have modulus below one in the finite atlas.

6.1 Why the raw zero residual is misleading

At $z = 1/\lambda_k$, the mathematically stable test is

$$\min_j |1 - z\lambda_j|.$$

Directly multiplying all factors in (4) can be unstable: other linear and exponential factors may become very large before multiplication by a factor of size 10^{-16}. The largest raw product residual in the audit is 6.51×10^4 even though the scale-free factor residual is 2.29×10^{-16}.

This is not evidence that the zero is absent. It is a conditioning warning, consistent with standard numerical practice for Fredholm determinants [1]. Later comparisons use logarithms on a disk satisfying $|z\lambda| < 1$.

7 What inversion repairs and what it does not

Inversion repairs the local-finiteness architecture at fixed noise: eigenvalues of a compact operator accumulate only at zero, so reciprocal zeros escape to infinity. It does not automatically repair a singular family in σ. As noise decreases, more resonances may remain away from zero, putting more reciprocal zeros in one fixed disk. A small-noise determinant family therefore needs:

1. local reciprocal-zero count stability;
2. omitted-factor control uniform on compact z-sets;
3. a moving-cloud or relative determinant if near-unit factors proliferate.

The next paper tests the first item on five fixed radii. Gate A remains open; the present determinant is nonself-adjoint and carries no established arithmetic trace formula or zeta divisor.

References

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